



INDIAN INSTITUTE OF TECHNOLOGY ROPAR

B.TECH. PROJECT REPORT

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Trajectory-Based Study of the Behavioural  
Dynamics of

*Turbatrix aceti*

Under Morphine Exposure and Post-Removal Conditions

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# Abstract

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Active matter consists of systems whose constituents continuously consume energy and convert it into mechanical motion. Biological microorganisms are important examples because their self-propelled motion occurs away from thermodynamic equilibrium and can be studied using tools from nonequilibrium statistical physics. In the present work, the swimming behaviour of the vinegar eel *Turbatrix aceti* is investigated under three experimental conditions: normal (control), morphine exposure, and after removal of morphine.

A bottom-glass Petri dish is used as the observation chamber. Image sequences are recorded using an AX200 camera at 125 frames per second for approximately 3 minutes per observation, with observations collected over a 12-day period. The recorded image sequences are processed using Fiji/ImageJ to generate videos and prepare the data for trajectory analysis. Individual nematodes are then tracked to obtain their two-dimensional positions as a function of time.

The present report describes the biological system, suspension preparation, experimental setup, image acquisition, image processing, and trajectory tracking. Quantitative analysis of the tracked trajectories is the next stage of the work. In particular, position and velocity correlations and mean square displacement (MSD) will be calculated and compared among the three experimental conditions. These measurements will provide a quantitative description of how morphine exposure and its subsequent removal influence the locomotory dynamics of *T. aceti*.

# Chapter 1

## Introduction

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### 1.1 Active Matter

Active matter is a class of nonequilibrium systems composed of constituents that consume energy and convert it into directed or self-generated motion. Unlike passive particles, whose motion is governed primarily by thermal fluctuations and externally imposed forces, active particles continuously produce mechanical motion by using energy from their environment or from an internal energy source. Examples include swimming microorganisms, motile cells, bacteria, synthetic microswimmers, cytoskeletal systems, and other self-propelled objects [1, 2].

A useful way to distinguish active matter from passive matter is through the concept of *self-propulsion*. An active constituent can generate motion without requiring a continuously applied external mechanical force in the direction of travel. Consequently, active systems operate out of equilibrium and can display dynamical behaviour that is not characteristic of conventional equilibrium materials [3, 2].

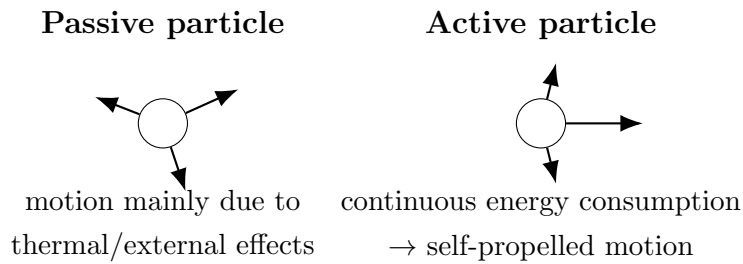


Figure 1.1: Schematic distinction between passive and active particles. Active particles continuously consume energy to generate motion.

The important point for the present study is that the motion of an active organism can be quantified from its trajectory. If the position of an organism at time  $t$  is represented by

$$\mathbf{r}(t) = [x(t), y(t)], \quad (1.1)$$

then its motion can be analysed using the time-dependent position and quantities derived from it.

## 1.2 Biological Active Matter

Biological systems provide natural examples of active matter. Microorganisms such as bacteria and other swimming organisms convert chemical energy into mechanical motion and thereby generate persistent, self-propelled trajectories. At larger scales, interactions between active constituents can also generate collective behaviour such as alignment, clustering, and coordinated motion [1, 4].

For the present work, the interest is not in reproducing a theoretical model of active matter but in using the active-matter viewpoint to quantify the locomotory dynamics of a biological swimmer. The experimentally recorded trajectory is therefore the primary observable.

## 1.3 *Turbatrix aceti* as the Experimental Organism

The organism used in this study is *Turbatrix aceti*, commonly known as the vinegar eel. It is a free-swimming nematode that is comparatively easy to culture and observe experimentally. Previous work has used *T. aceti* as a model system for experimental studies of active biological suspensions [5].

Reported characteristics of *T. aceti* include a body length of about 1 mm and a diameter of about 22  $\mu\text{m}$ . The organism propels itself through swimming motion and has been reported to have a characteristic swimming speed of approximately  $0.4 \text{ mm s}^{-1}$ . These properties make it convenient for optical observation and trajectory-based measurements [5].

In this project, the purpose of using *T. aceti* is to investigate whether its observable locomotory dynamics change under morphine exposure and after subsequent removal of the drug.

## Chapter 2

# Problem Statement

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The objective of this project is to investigate the behavioural dynamics of *Turbatrix aceti* under different chemical conditions.

Three experimental states are considered:

1. **Normal/control condition:** the baseline swimming behaviour of *T. aceti* is recorded without morphine exposure.
2. **Morphine-exposed condition:** the organisms are observed after exposure to morphine according to the experimental protocol.
3. **Post-removal condition:** after the morphine-containing condition is removed/replaced according to the experimental protocol, the subsequent swimming behaviour is recorded.

The central problem can therefore be stated as:

**How does morphine exposure affect the locomotory dynamics of *Turbatrix aceti*, and how does the observed motion change after removal of the drug?**

The study is designed as a trajectory-based investigation. Instead of relying only on visual observations, the recorded motion is converted into measurable position and velocity data. These data will subsequently be used to calculate correlation functions and mean square displacement.

## Chapter 3

# Methodology

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### 3.1 Vinegar-Eel Suspension

The biological material used in the experiments was *Turbatrix aceti* (vinegar eel). The culture preparation followed the general procedure reported for *T. aceti* suspensions [5].

The culture medium was prepared using food-grade apple cider vinegar and distilled water. A portion of the starter culture containing *T. aceti* was introduced into the prepared culture medium, followed by the addition of peeled apple pieces. The culture container was covered while maintaining aeration. Over approximately 10–15 days, a dense population of vinegar eels developed.

For the literature-based preparation procedure, the dense culture can be filtered and the collected nematodes washed and concentrated before use in experiments. The schematic supplied for this work is shown in Fig. 3.1. The exact concentrations and preparation quantities used in the present laboratory experiment should be reported from the experimental notebook rather than inferred from the literature.

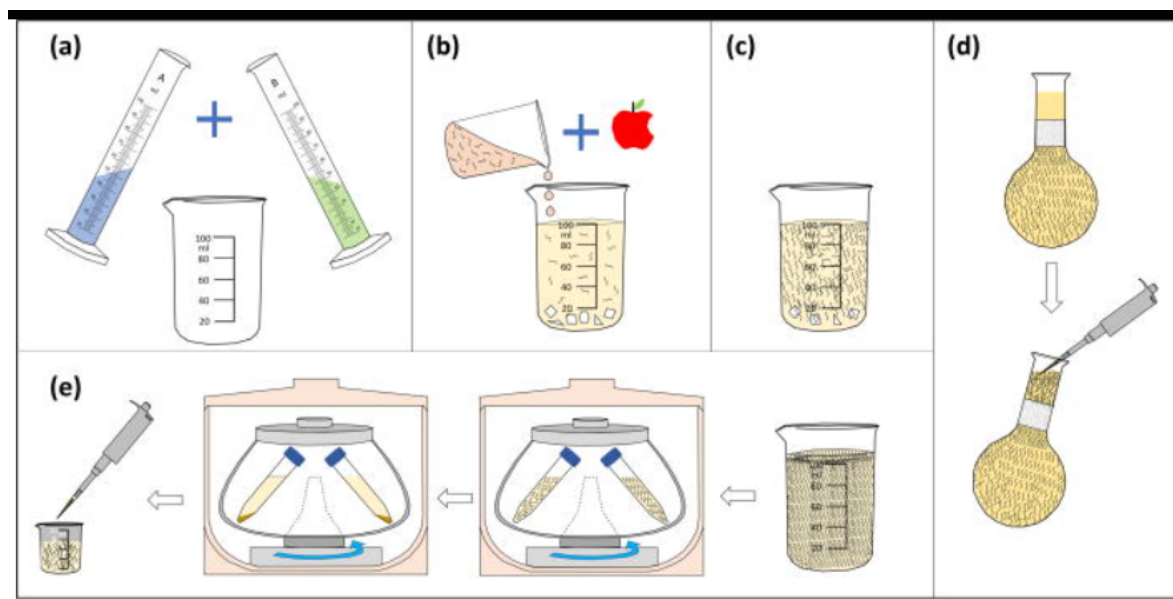


Figure 3.1: Schematic of the preparation and concentration of a *Turbatrix aceti* suspension. The workflow shows preparation of the vinegar-based culture medium, growth of the culture, filtration, and collection/concentration of the nematodes. Adapted from the preparation procedure reported by Ali *et al.* [5].

## 3.2 Experimental Conditions

The behavioural study was performed under three conditions:

Table 3.1: Experimental conditions considered in the study.

| Condition         | Purpose                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------|
| Normal / Control  | Establish the baseline locomotory behaviour of <i>T. aceti</i> .                              |
| Morphine exposure | Record the locomotory behaviour under the specified morphine treatment.                       |
| Post-removal      | Observe the locomotory behaviour after removal/replacement of the morphine-containing medium. |

## 3.3 Experimental Setup

A bottom-glass Petri dish was used as the observation chamber. The glass bottom provided an optically accessible surface for recording the movement of the organisms.

The main imaging parameters used in the present experiment are summarized in Table 3.2.

Table 3.2: Image-acquisition parameters used in the experiment.

| Parameter           | Experimental value                          |
|---------------------|---------------------------------------------|
| Observation chamber | Bottom-glass Petri dish                     |
| Camera              | AX200                                       |
| Frame rate          | 125 fps                                     |
| Recording duration  | Approximately 3 min per observation         |
| Observation period  | 12 days                                     |
| Primary data        | Image sequences of swimming <i>T. aceti</i> |

### 3.4 Image Acquisition

Images of the nematode suspension were acquired using the AX200 camera at a frame rate of 125 frames per second. Each observation was recorded for approximately three minutes. Observations were collected over a period of twelve days under the experimental conditions described above.

The high frame rate provides a time-resolved record of the swimming organisms, which is necessary for subsequently extracting their positions and calculating velocity-related quantities.

### 3.5 Image Processing Using Fiji/ImageJ

The recorded image sequences were processed using Fiji/ImageJ. The image sequence was imported into Fiji and converted into a video/stack suitable for visualization and subsequent tracking.

A representative raw image from the experiment is shown in Fig. 3.2. It illustrates the optical field containing the vinegar eels before trajectory extraction and provides a visual example of the image data imported into Fiji/ImageJ.

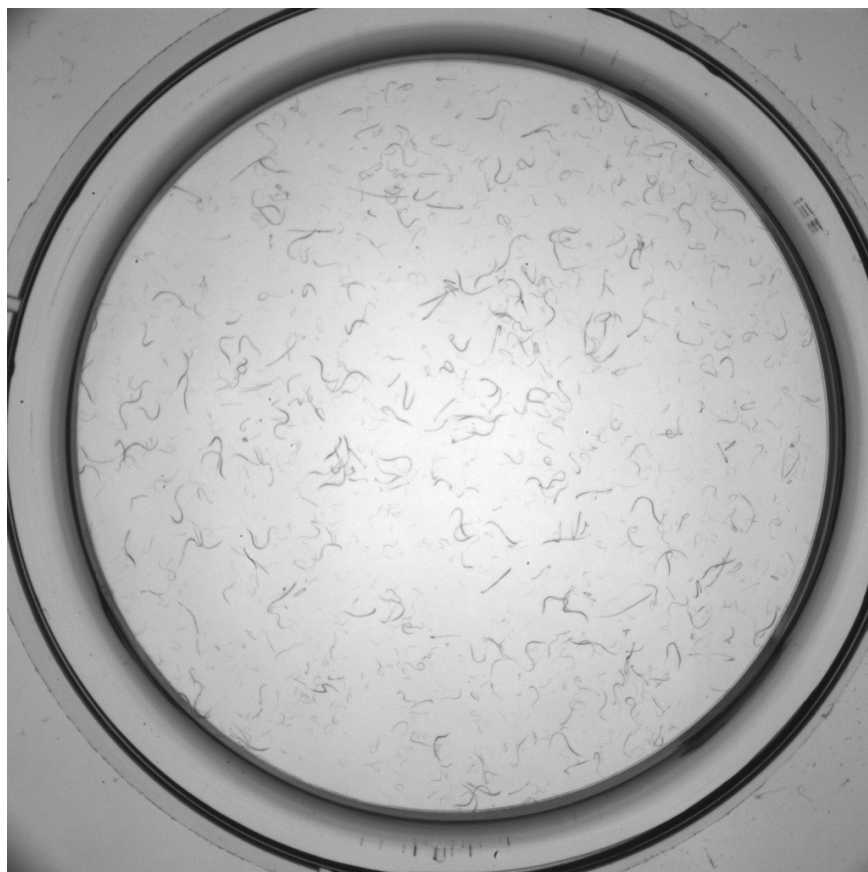


Figure 3.2: Representative raw image of the *Turbatrix aceti* suspension used to illustrate the image-processing stage in Fiji/ImageJ. The image sequence is imported into Fiji/ImageJ and prepared for subsequent visualization and trajectory tracking.

### 3.6 Trajectory Tracking

After image processing, individual organisms were tracked using Fiji/ImageJ and the TrackMate workflow. Tracking provides the two-dimensional position of an identified organism at successive time points,

$$\mathbf{r}(t) = [x(t), y(t)]. \quad (3.1)$$

The successive positions are joined to obtain an individual trajectory. The trajectory data obtained from tracking constitute the primary dataset for the next stage of the project.

## Chapter 4

# Results and Discussion

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### 4.1 Day-0 Experimental Observations

Day 0 imaging was used to document the initial visual state of the suspension under the three experimental conditions considered in the study. The images were acquired from the bottom-glass Petri-dish observation chamber and show the spatial distribution of *Turbatrix aceti* within the field of view. These images are presented as experimental observations rather than as quantitative measurements; the subsequent trajectory analysis will provide the numerical basis for comparison.

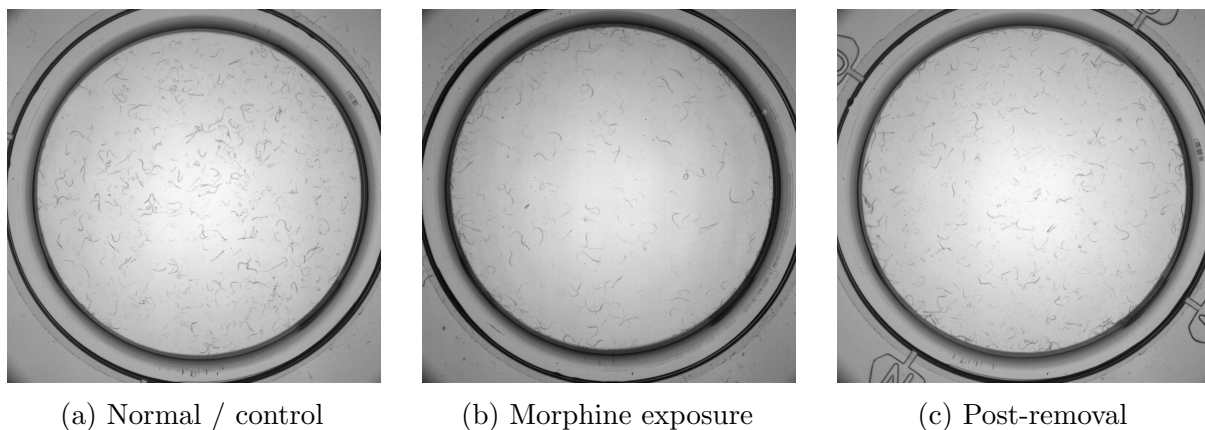


Figure 4.1: Day-0 experimental images for the three study conditions: normal / control, morphine exposure, and post-removal. The images provide a visual record of the observed nematode suspension before quantitative trajectory analysis.

The Day-0 images show that the nematodes are distributed throughout the observation chamber in all three recorded conditions. Differences in apparent density or spatial distribution should not be interpreted quantitatively from these images alone because illumination, focus, overlap between organisms, and image-processing conditions can influence visual appearance. A quantitative comparison will therefore be based on the tracked coordinates obtained from the corresponding image sequences.

## 4.2 Trajectory-Tracking Result

The current stage of the work has established the imaging and trajectory tracking workflow for *Turbatrix aceti*. The recorded image sequences can be converted into videos in Fiji/ImageJ, and individual organisms can be followed across successive frames to obtain two-dimensional trajectories.

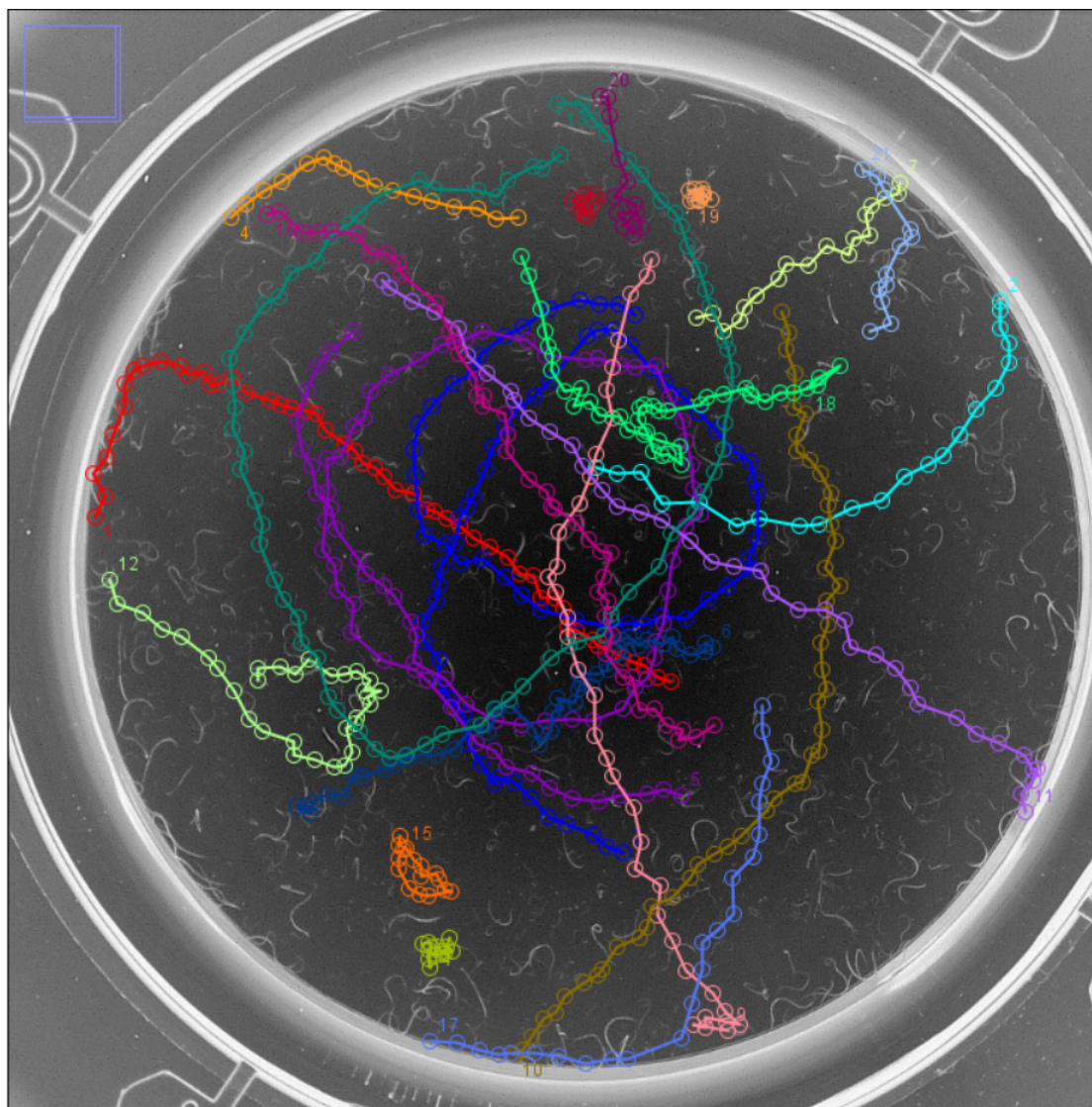


Figure 4.2: Representative trajectory-tracking result obtained from the recorded *Turbatrix aceti* image sequence. Different colours correspond to different tracked organisms and illustrate the spatial paths extracted from the image sequence.

Figure 4.2 demonstrates that multiple individual organisms can be tracked simultaneously. The resulting trajectories provide the necessary spatial information for quantitative analysis of position, velocity, correlation functions, and mean square displacement.

At this stage, no quantitative conclusion about the effect of morphine is drawn from

the trajectory image alone. The next stage is to calculate the statistical and dynamical quantities from the tracked coordinates. This is important because visual differences in trajectories should be supported by quantitative measurements before interpreting a change in locomotory behaviour.

### 4.3 Comparison Framework

The quantitative comparison will be performed using the same analysis procedure for the three experimental states:



The objective is to determine whether the statistical properties of the trajectories differ between these conditions.

## Chapter 5

# Future Work

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The trajectory extraction completed in the present stage provides the basis for quantitative active-matter analysis. The following measurements will be calculated from the tracked position data.

### 5.1 Position

For each tracked organism, the position will be represented as

$$\mathbf{r}(t) = [x(t), y(t)]. \quad (5.1)$$

The temporal evolution of  $x(t)$  and  $y(t)$  will be used as the starting point for the subsequent analysis.

### 5.2 Velocity

The instantaneous velocity will be obtained from the trajectory. For discrete data, it can be approximated as

$$\mathbf{v}(t) \approx \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}. \quad (5.2)$$

The corresponding speed is

$$v(t) = |\mathbf{v}(t)|. \quad (5.3)$$

Velocity statistics will be compared among the control, morphine-exposed and post-removal conditions.

### 5.3 Position Correlation

The temporal correlation of particle position will be evaluated to quantify how strongly the position at one time is related to the position at a later time. A general position-correlation function can be written as

$$C_r(\tau) = \langle \mathbf{r}(t) \cdot \mathbf{r}(t + \tau) \rangle, \quad (5.4)$$

with an appropriate normalization applied when required by the analysis.

The position correlation will provide information about the temporal persistence of the trajectory.

### 5.4 Velocity Correlation

The velocity correlation function will be used to quantify the temporal persistence of swimming motion. It can be written as

$$C_v(\tau) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle. \quad (5.5)$$

A normalized form can be expressed as

$$C_v^N(\tau) = \frac{\langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle}{\langle v^2(t) \rangle}. \quad (5.6)$$

Velocity correlations are a standard measure of temporal dynamics in active particle systems. The supplied theoretical reference on inertial self-propelled particles explicitly treats time-dependent velocity correlations and their characteristic decay times [6].

### 5.5 Mean Square Displacement

The mean square displacement (MSD) will be calculated from the tracked trajectories as

$$MSD(\tau) = \langle |\mathbf{r}(t + \tau) - \mathbf{r}(t)|^2 \rangle. \quad (5.7)$$

MSD describes the average spatial displacement of the organisms over a time lag  $\tau$ . It is particularly useful for distinguishing different regimes of motion. The supplied reference paper derives MSD from the temporal velocity correlation and discusses its transition from short-time ballistic behaviour to long-time diffusive behaviour [6].

For a simple diffusive regime,

$$MSD(\tau) \propto \tau, \quad (5.8)$$

whereas a ballistic regime has

$$MSD(\tau) \propto \tau^2. \quad (5.9)$$

The MSD curves obtained for the three experimental conditions will be compared to determine whether morphine exposure changes the spatial dynamics of the organisms.

## 5.6 Overall Quantitative Comparison

The final quantitative analysis will therefore focus on:

1. tracked position  $\mathbf{r}(t)$ ;
2. velocity  $\mathbf{v}(t)$ ;
3. position correlation;
4. velocity correlation;
5. mean square displacement (MSD).

These quantities will be calculated consistently for the control, morphine exposure, and post-removal conditions. The resulting curves and statistical parameters will then be compared to identify changes in locomotory dynamics.

## Chapter 6

# Conclusion

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The present work establishes an experimental and image-analysis framework for studying the locomotory behaviour of *Turbatrix aceti* as an active biological system. The organisms are observed under normal conditions, morphine exposure, and after removal of the morphine-containing condition.

Image sequences are acquired at 125 fps for approximately three minutes and collected over twelve days. Fiji/ImageJ is used to process the image sequences, and trajectory tracking is performed to obtain the positions of individual organisms.

The current trajectory results demonstrate that the recorded image sequences can be converted into trackable two-dimensional trajectories. The next stage of the project will transform these trajectories into quantitative measures of position, velocity, position correlation, velocity correlation, and mean square displacement. These measurements will provide a quantitative basis for comparing the locomotory dynamics across the three experimental conditions.

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